

Prediction of Exchange Bias for Magnetic Heterostructure Nanoparticles with Machine Learning

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ABSTRACT: Exchange bias is essential for the stability and control of nanoparticles' magnetic properties for their application as rare-earth-free permanent magnet, magnetic storage, magnetic hyperthermia, and catalysis. Core-shell structures of magnetic bimagnetic particles have garnered increasing interest due to their larger coercive and exchange bias fields, tunable blocking temperatures, and enhanced Neel temperature. However, the design approach of nanoparticles with exchange bias using a computational method has a high computational cost and offers limited efficiency in predicting complex core-shell nanoparticle systems. Machine learning (ML) predictions provide a transformative approach to the design and optimization of materials with desirable exchange bias (EB) properties by offering rapid and precise evaluations of material compositions and configurations. This study addresses this gap by developing an ML model for the prediction of EB field in magnetic nanoparticles, which provide a fast and effective alternative for traditional computational methods. Hence, comparative analysis of ML models, including the Kolmogorov-Arnold Network (KAN), is for predicting EB in core-shell and heterostructure nanoparticles. Among the predictive models, XGBoost demonstrated superior performance, achieving R^2 values of 0.74 and 0.75 on the test and validation data sets, respectively. KAN showed reduced generalization power with the R^2 test of 0.67 but was more accurate in predicting high values of the EB. The Shapley additive explanation (SHAP) analysis revealed unexpected dependencies between nanoparticle properties and magnetic behavior, offering new insights for optimizing a material design. These findings are highly relevant for developing materials for rare-earth-free permanent magnets, magnetic storage, magnetic hyperthermia, and catalysis, where precise control of magnetic properties is crucial.



INTRODUCTION

Magnetic heterostructures have attracted tremendous attention due to their unique exchange coupling with promising properties for emerging multiferroic¹ and spintronic^{2–4} technologies compared with traditional, single-phase materials. These materials are categorized based on their core and shell structures, which exhibit different types of magnetic ordering, such as ferromagnetism (FM), ferrimagnetism (FiM), anti-FM (AFM), spin-glass behavior (SG), spin disorder (SD), and paramagnetism (PM).⁵ EB is generally considered as a fundamental magnetic interaction between FM and AFM heterogeneous structures, leading to a shift of the hysteresis loop along the field axis after field cooling below the Néel temperature (T_N).^{6,7} This effect is highlighted by a horizontal shift of the hysteresis loop as well as an increase in the coercivity.⁸

Although the EB effect is well understood in thin films,⁹ bimagnetic nanoparticles are excellent candidates for creating exchange-coupled magnetic materials. This is because the dimensions of the two magnetic phases can be adjusted through the core and shell diameters, which is advantageous for studying spin coupling across interfaces and for tuning the magnetic properties of the nanoparticles. Core/shell structures offer an additional freedom to control interfacial effects, where

the symmetry is significantly different (three-dimensional compared to two-dimensional in planar films), and finite size effects can also play a role in influencing the magnetic behavior.^{10–12} Moreover, despite the extensively studied the FM/AFM core-shell nanoparticles, the inverted structures (AFM/FM and AFM/FiM) have an increasing interest due to larger coercive and EB field, tunable blocking temperatures, and enhanced Neel temperature.^{13–16} From a fundamental point of view, such magnetic core-shell systems can become a meaningful toolbox for understanding and controlling EB effects, and they have already been found in practical application as rare-free permanent magnets,^{14,17} magnetic memory,^{18–20} magnetic hyperthermia,¹⁵ and in magnetic field-mediated electrocatalysis.²¹

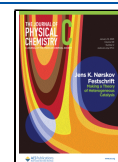
The mechanism of EB is primarily connected to interfacial interactions between FM and AFM components, mainly from pinned, uncompensated magnetic moments in the AFM layer.

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Recent studies also highlight the impact of nonmagnetic impurities and crystallographic defects on EB.^{9,22} Generally, the EB field depends on material composition, interface quality, cooling field conditions, and structural parameters such as exchange coupling, surface defects, crystalline structure, and anisotropy and could be tuned by rational design of nanoparticles.^{7,23–25} Stamps²⁶ noted that strong negative EB fields arise from pinned moments interacting with the FM lattice. Avila-Gutierrez et al. demonstrated a significant role of crystallinity and optimal core/shell ratio in EB field for Co/CoO nanoparticles²⁴. Muzzi et al.²⁸ highlight defect engineering as a promising tool to induce the EB effect in ferrites. Such examples showed the variety of design strategies to control EB as well as the number of parameters, which should be considered for rational design.

Computational methods such as micromagnetic simulations, Monte Carlo simulations, and *ab initio* calculations assist in designing materials with EB.^{29,30} Micromagnetic simulations, performed using tools such as OOMMF, provide insights into nanoscale magnetic behavior and phenomena like magnetization reversal.^{31,32} While density functional theory (DFT) is effective in predicting intrinsic magnetic properties, such as magnetization and exchange interactions, it struggles to accurately describe extrinsic properties like coercivity. On the other hand, micromagnetic simulations, although useful for modeling coercivity, often overestimate their values due to limitations such as Brown's paradox.³³ Micromagnetic simulation for the simulation of EB further complicates the assignment of a single magnetic moment per unit cell, which limits its ability to accurately simulate the complex magnetic ordering present in AFM or FiM materials. This limitation is particularly significant when attempting to model exchange coupling at the FM/AFM interface. Additionally, the importance of accounting for thermal and quantum fluctuations significantly increases the number of required features, thereby escalating the computational costs.⁴³ While atomistic simulation (e.g., Vampire) could capture the magnetic ordering and thermal effects, enabling a more accurate modeling of EB and interfacial phenomena in small core-shell particles, it suffers from high computation cost and limited scalability in the case of larger particle sizes. Moreover, to gain a more comprehensive understanding of the magnetic phenomena even within a single material, it is often necessary to use multiple simulations or calculations. For instance, when studying single-layer or bilayer materials, a combination of computational methods and Monte Carlo simulations is employed to explore the interrelations between magnetic parameters and their mutual influences.^{44,45} This, in turn, greatly increases the time and resource demands in material research and the development of new materials.

ML methods are increasingly used to enhance magnetic property predictions by integrating experimental data, DFT results, and micromagnetic simulations.³⁴ Traditional techniques are limited by high computational costs and challenges in accounting for quantum effects and varied sample parameters.^{27,35–38} Recent ML applications include predicting Curie temperatures for rare-earth intermetallic systems,³⁹ force fields of metallic nanoparticles based on morphology^{40,41}, and magnetic remanence of Sm–Co magnet alloys.⁴¹ Other studies have also used two-dimensional material databases to predict net magnetic moments and classify materials as magnetic or nonmagnetic.⁴² However, ML applications for predicting EB in

nanomaterials have been less explored, revealing a significant research gap.

This study addresses this gap with the development of an ML model for the prediction of EB field in magnetic nanoparticles, which provide a fast and effective alternative for traditional computational methods. Hence, a comparative analysis of ML models, including the KAN, for predicting EB in core-shell and heterostructure nanoparticles was conducted. The XGBoost model achieved the highest R^2 scores on training, test, and validation sets (0.75, 0.74, and 0.75, respectively), outperforming KAN on the test set by 7%. KAN demonstrated a comparable performance as assessed by cross-validation and test metrics. Interestingly, KAN featured a more balanced decision-making with evenly distributed feature importance, resulting in better capturing of large EB shifts, crucial for detailed material analyses. Feature importance analysis identified coercivity, anisotropy field, and the Néel-to-blocking temperature ratio as being critical for EB shifts. ML models also effectively capture a temperature-dependent behavior of the EB field and have a strong agreement with experimental results. Therefore, our approach combining various ML techniques provides a more comprehensive understanding of magnetic phenomena and opens up new possibilities for an advanced material design.

METHODS

Data Collection. The data set was meticulously assembled from various scientific sources, capturing a comprehensive range of descriptive data, including physicochemical properties, EB shift measurements, measurement temperatures, applied magnetic fields, and resulting magnetization. The data were manually extracted and verified.

To ensure the inclusion of relevant studies, we conducted a thorough search on Scopus using key terms such as “core-shell AFM/FM nanoparticles”, “EB shift”, “FM/AFM core-shell”, and “magnetic heterostructure”. This methodical approach allowed us to compile a robust database of magnetic core-shell nanoparticles, providing a solid foundation for subsequent analysis. The data sources with a description of the training effect of EB were filtered and excluded from the database. All the articles from which the data were extracted to create the unified data set are listed in Table S1. The data collected on nanoparticle composition revealed a broad diversity. Although the primary objective was to analyze core-shell AFM/FM or FM/AFM nanoparticles, the final data set included various materials.

Descriptors and Data Processing. The documented parameters included nanoparticle geometric features (size, shape, sphericity, geometric anisotropy, and area-to-volume ratio), magnetic characterization (exchange stiffness coefficient, Curie temperature, Neel temperature, blocking temperature, coercivity, saturation and remanent magnetization, EB shift, exchange shift direction, vertical shift, and vertical shift direction), SQUID experimental parameters (maximum field range, magnetic field strength, and experimental temperature), physicochemical properties of nanoparticles, electronic and crystal parameters for core and shell of nanoparticles (core b lattice parameter, core alpha angle, core beta angle, core gamma angle, shell b lattice parameter, shell alpha angle, shell beta angle, shell gamma angle, magnetic ion ratio, core lattice ratio c to a, Pauling electronegativity of core, Pauling electronegativity of shell, valence electron concentration of core, work function of core, valence electron concentration of

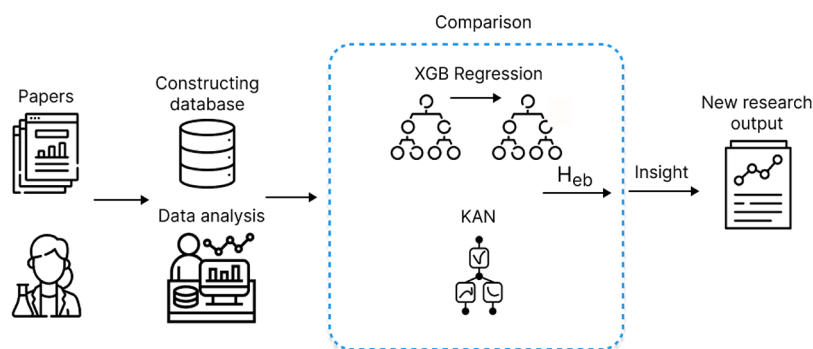


Figure 1. Research pipeline for prediction of EB in magnetic heterostructure nanoparticles using classical ML and KAN.

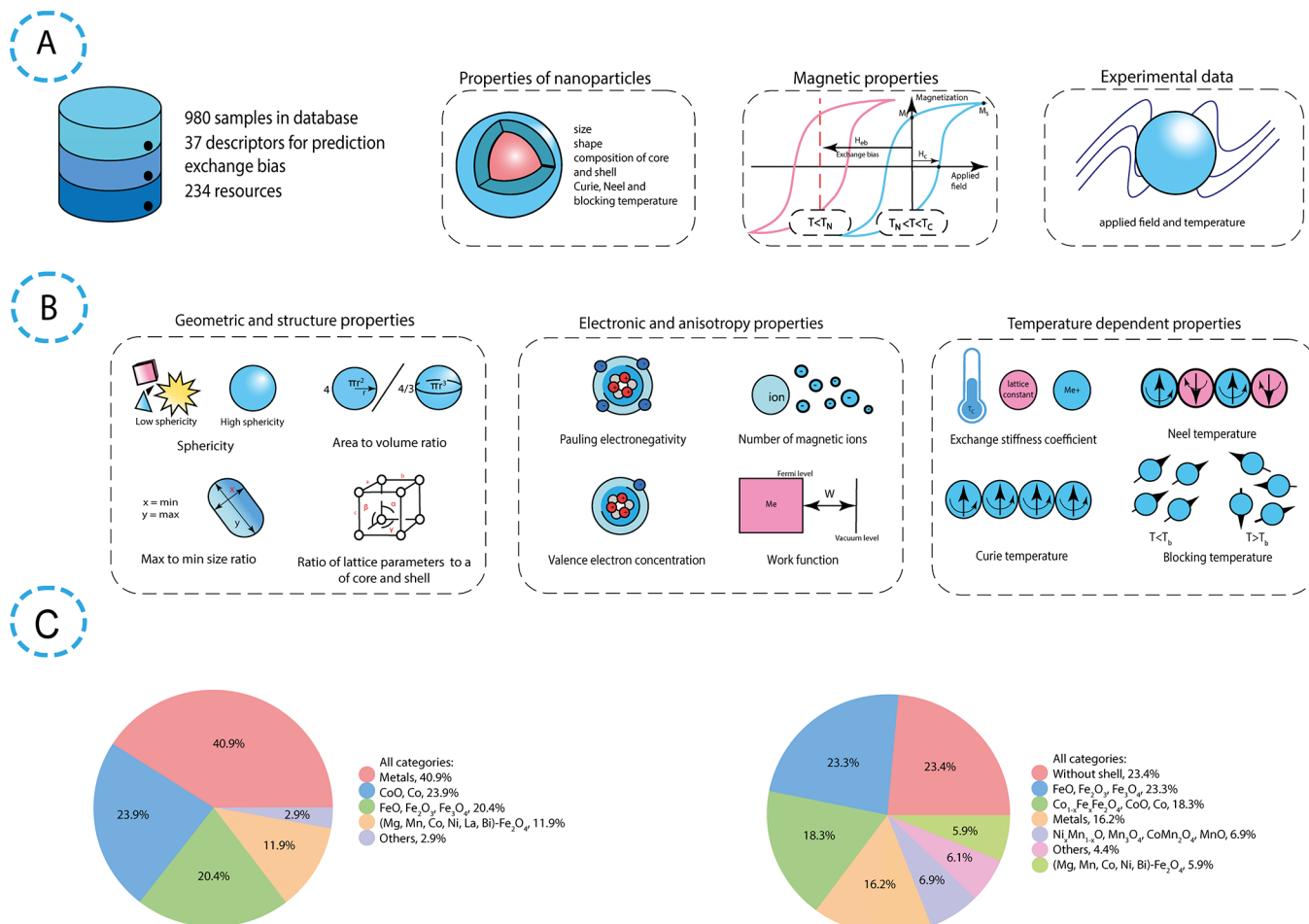


Figure 2. Overview of the database of heterostructured nanoparticles with EB. (A) Summary of database composition, detailing the parameters and source compositions extracted from scientific articles; (B) feature engineering process, showing the extraction of relevant features and generation of additional parameters; (C) statistical analysis, featuring a pie chart of core and shell material composition distributions.

shell, work function of shell, Pauling electronegativity difference of core, valence electron concentration difference of core, work function difference of core, and Pauling electronegativity).

The abovementioned descriptors were obtained after a correlation matrix analysis. Pearson's correlation coefficient was used as the measure of linear association or statistical relationship between two variables. Parameters that had strong correlations were removed because these correlations decrease the performance of ML models.

For the characterization of the collected data, violin plots were generated to display the distribution of all parameters (Figure S4). These plots revealed asymmetries in several variables, highlighting the nonuniform distribution of the data. Additionally, Table S2 provides a comprehensive summary of the descriptive statistics and a brief description of each variable.

For missing data, we used Springer Materials⁵² as an additional source. We applied k-nearest neighbor (KNN) imputation for missing data up to 15%, covering the Néel temperature, Curie temperature, saturation magnetization,

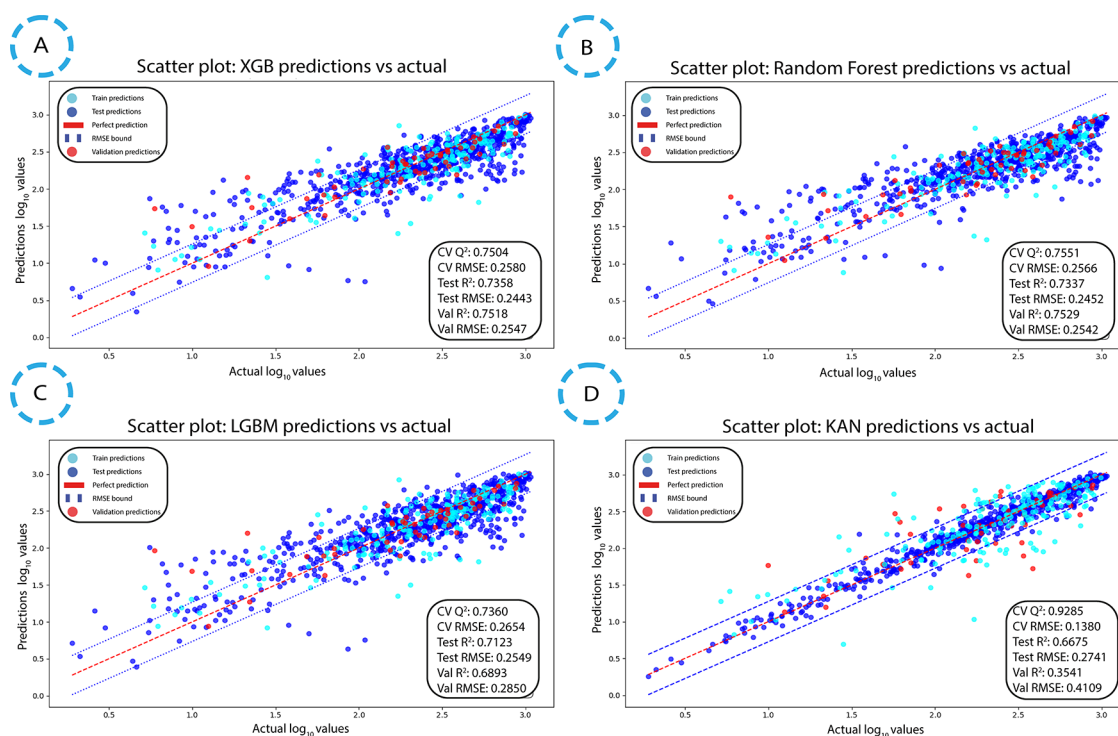


Figure 3. Comparison of the best model predictions and performances for the four top-performing models: (A) XGBoost, (B) random forest, (C) LGBM, and (D) KAN is shown predicting the EB.

remanence magnetization, vertical exchange shift, and measurement conditions. The feature exhibiting approximately 50% or more missing values was entirely excluded from the data set. Anomalous entries were removed using the interquartile range (IQR) method, resulting in a more reliable data set, with final feature distributions shown in Figure S4.

The final data set, used for predicting EB and coercive force, consisted of 980 samples characterized by 37 descriptors (detailed in Table S2). An independent validation data set of 51 samples from recent studies (2024) was included for unbiased model performance assessment.

EB Prediction. The data set was split into training and testing sets at an 80 to 20% ratio. The target variable was log-transformed. We employed a fivefold cross-validation as a robust method for evaluating model performance and minimizing overfitting risks. Hyperparameter tuning for KAN was conducted using Optuna (Table S3, Supporting Information). We optimized and compared the performance of the four best ML models for each regression problem (Table S3, Supporting Information). Final model selection was based on mean R^2 and RMSE for cross-validation and R^2 and RMSE for test and validation samples, ensuring accurate and generalizable predictions for nanoparticle magnetic properties. For interpretation of predicted results of ML models, SHAP values were used. Using them, the importance of each descriptor and its influence on the predicted value can be analyzed.

RESULTS AND DISCUSSION

The graphical overview of this work is given by Figure 1. We conducted a comprehensive review of experimental articles on EB in magnetic nanomaterials (Table S1). That led to the creation of a data set with 1400 entries detailing structural

parameters of particles, including metals, doped oxides, complex oxides, intermetallics, and core-shell structures.

The documented parameters include core-shell structures, particle shape and size, magnetic field, sphericity, space group, particle anisotropy constants, Curie and Néel temperatures, saturation magnetization, remanent magnetization, coercive force, EB magnitude and direction, measurement temperature, and more (see ESI, Table S2). Additionally, the exchange stiffness coefficient (A_{ex}), crucial for magnetic phenomena like skyrmion formation and magnetic domain wall width,^{46–50} was included to indicate the spin interaction strength.

We characterized the nanoparticles using 37 parameters (Table S2 and Figure 2). Key parameters such as particle size, shape, measurement temperature, field, saturation magnetization, remanence magnetization, coercivity, and EB were manually extracted. Due to incomplete reporting, descriptors like the core-shell quantitative ratio, exchange constant, and interdomain coupling were excluded because missing values exceeded 50%, as retaining such incomplete data would have compromised the model performance.

To enhance the characterization and prevent overfitting, we applied advanced feature engineering. We transformed geometric parameters into ratios, creating features such as sphericity, minimum-to-maximum dimension ratio, and surface area-to-volume ratio. We also computed the ratio $\frac{T_N - T_B}{T_N}$ to assess temperature ranges where materials maintain anti-ferromagnetic properties while exhibiting a superparamagnetic behavior.^{51,52} Features related to crystal lattice parameters were generated using Pymatgen, enhancing magnetic behavior analyses. We constructed a correlation matrix to eliminate redundant parameters, with initial and final matrices shown in Figures S4 and S5. The correlation analysis confirmed linear relationships between new descriptors (Figure 3 and Figure S5). For missing data, we used Springer Materials⁵² as an

additional source. We applied KNN imputation for missing data up to 15%, covering Néel temperature, Curie temperature, saturation magnetization, remanence magnetization, vertical exchange shift, and measurement conditions. The final data set, used for predicting EB and coercive force, consisted of 980 samples characterized by 37 descriptors (detailed in Table S2).

We employed various tree-based ensemble ML models for prediction due to their effectiveness with tabular data sets and the ability to capture complex nonlinear relationships.⁵³ Tree-based models handle outliers and missing values better than do deep learning models.⁵⁴ Linear regression, logistic regression, and support vector machines did not perform well. Ensemble models such as extra trees and random forest showed promising results, while bagging, gradient boosting, hist gradient boosting, XGBoost, and LightGBM (LGBM) performed best, managing both numerical and categorical features related to magnetic parameters. We also explored the recently proposed KAN architecture as a multilayer perceptron (MLP) alternative with learnable activation functions.^{55,56} KANs have already been successfully applied to some scientific domains,⁵⁷ so we wondered about the performance of KAN in our task and its ability to capture dependencies in magnetic material properties.

As the EB effect was originally found as a coercivity effect, we also studied the robustness of the developed models for coercivity prediction. The results of coercivity prediction by tree-based models and KAN are given on Figure S6. The Bagging model proved to be more suitable in our case, with test metrics of $R^2 = 0.73$ and RMSE = 0.25, demonstrating the highest stability and overall performance, as shown in Table S3.

Similar results for the EB prediction are shown in Figure 3. Table 1 further details models' performance, providing cross-

Table 1. Performance Comparison of Models Predicting EB^a

model /metric	train set		test set		validation set	
	CV Q^2	CV RMSE	R^2	RMSE	R^2	RMSE
random forest	0.76	0.26	0.73	0.25	0.75	0.25
LGBM	0.74	0.27	0.71	0.25	0.69	0.29
XGB	0.75	0.26	0.74	0.24	0.75	0.25
KAN	0.93	0.14	0.67	0.27	0.35	0.41

^aFor the train set, fivefold cross-validation (CV) metrics are given.

validation, test, and validation metrics. The XGB model demonstrated strong performance in predicting EB, achieving a fivefold cross-validation RMSE of 0.26 and Q^2 of 0.75, with a test RMSE of 0.24 and R^2 of 0.74.

In contrast, KAN demonstrated an even stronger performance on the train set with the CV RMSE of 0.14 and Q^2 of 0.93 but achieved only RMSE of 0.27 and R^2 of 0.67 on the test set. These empirical results indicate the comparable predictive power of KAN with a slightly reduced generalizability compared to the best ensemble models. However, a notable distinction emerged from considering the predictions of high values of the EB. We found that KAN is particularly adept at modeling such subtle variations in the EB (Figure S5B), which seem to be overlooked by the other ML algorithms. This finding underscores the importance of selecting models based on not only the average accuracy but also their ability to capture specific patterns within the data, especially when dealing with complex physical phenomena like EB.

Ensuring model interpretability remains a significant challenge in applying ML to scientific problems, particularly in trying to understand and explain the decision-making process of complex models. To address this, we employed SHAP values to evaluate feature importance across various ML models. SHAP values, grounded in game theory, provide a robust framework for interpreting model outputs by attributing important scores to each feature for individual predictions. This approach enhances our understanding of feature contributions to model predictions, as illustrated on 4.

In our SHAP analysis of the random forest and XGBoost models for predicting EB, several dependencies emerged, shedding light on the factors influencing this magnetic phenomenon. Coercivity consistently emerged as the most significant predictor in both models, aligning with the established principles in magnetism. Higher coercivity, reflecting a material's resistance to demagnetization, correlates with stronger magnetic anisotropy; however, while coercivity and EB are often observed to coexist, coercivity does not directly enhance EB.⁵⁸ The mechanisms of coercivity increase associated with interface AF spin clusters, which increase an energy barrier for magnetization reversal.^{58,59} The anisotropy field (ratio of magnetic anisotropy to saturation magnetization) and the demagnetizing factor related to material geometry also impact EB. Although specific geometric ratios of lattice parameters may not directly correlate with EB, the overall shape significantly affects the EB. This highlights the importance of incorporating comprehensive geometric parameters into predictive models to capture the interplay among magnetic anisotropy, material morphology, and EB.

The dependence of target variables on temperature has been investigated in several studies,^{57–59,60–62} which our models corroborate, as seen on Figure 4. The importance of saturation magnetization and remanence magnetization was anticipated. The M_r/M_s ratio indicates a material's magnetic strength and coercive behavior, with higher ratios associated with stronger magnetic anisotropy and increased EB. Materials with higher remanence and saturation magnetization exhibit stronger EB effects, validating the link between robust magnetic properties and EB.^{63,64} Temperature-related variables influence the EB in predictable ways. Specifically, a larger ratio $\frac{T_N - T_B}{T_N}$ corresponds

to a weaker EB field, as the blocking temperature approaches or exceeds the Néel temperature. When $T_B > T_N$, the antiferromagnetic order responsible for pinning ferromagnetic domains is lost, leading to magnetic domain reorientation and disappearance of EB. Conversely, a higher Néel temperature relative to the blocking temperature enhances EB by preserving the antiferromagnetic order over a wider temperature range. This relationship highlights the importance of selecting materials where T_N significantly exceeds T_B to ensure robust magnetic stability and strong EB.⁶⁵

The Pauling electronegativity and valence electron concentration (VEC) of the core material have emerged as moderate predictors with an influence on EB field prediction. Although these descriptors have not directly been extensively studied in relation to EB,^{66–68} they might serve as indirect attributes for complex interfacial phenomena that influence exchange coupling. Specifically, Pauling electronegativity may reflect differences in charge redistribution, bonding strength, or orbital hybridization at the FM/AFM interface. In addition to electronegativity, the VEC plays a significant role in determining the density of states at the Fermi level, the

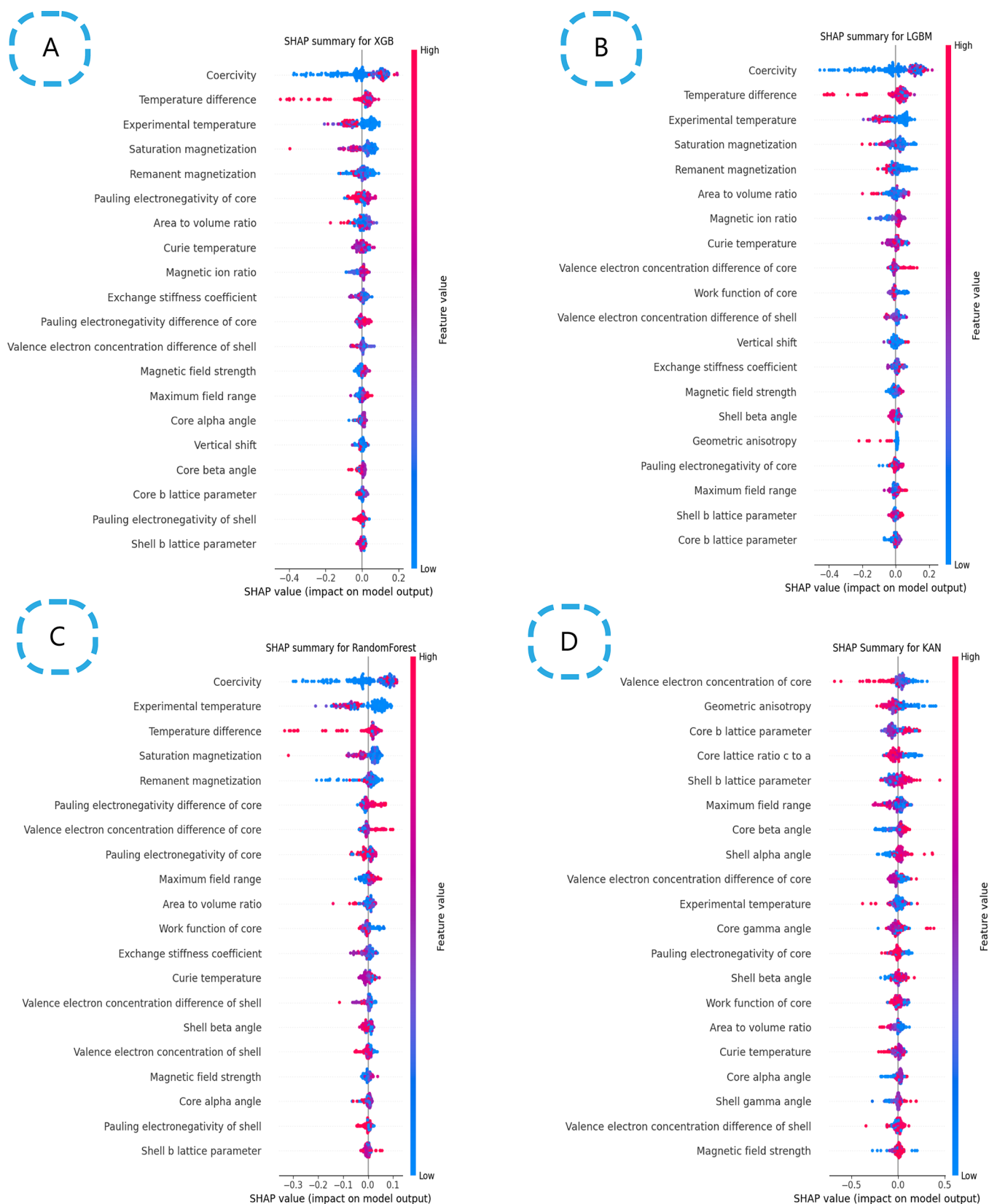


Figure 4. SHAP summary plot, showcasing the key features driving the model's decisions for top-performing models for prediction EB: (A) XGBoost, (B) LGBM, (C) random forest, and (D) KAN.

availability of unpaired electrons, and the spin polarization in both FM and AFM layers. Epitaxial strain, local symmetry breaking, and charge transfer at these interfaces can

significantly affect EB.⁶⁹ The interaction between core and shell regions is also pivotal, with stronger interfacial coupling typically enhancing EB.⁷⁰

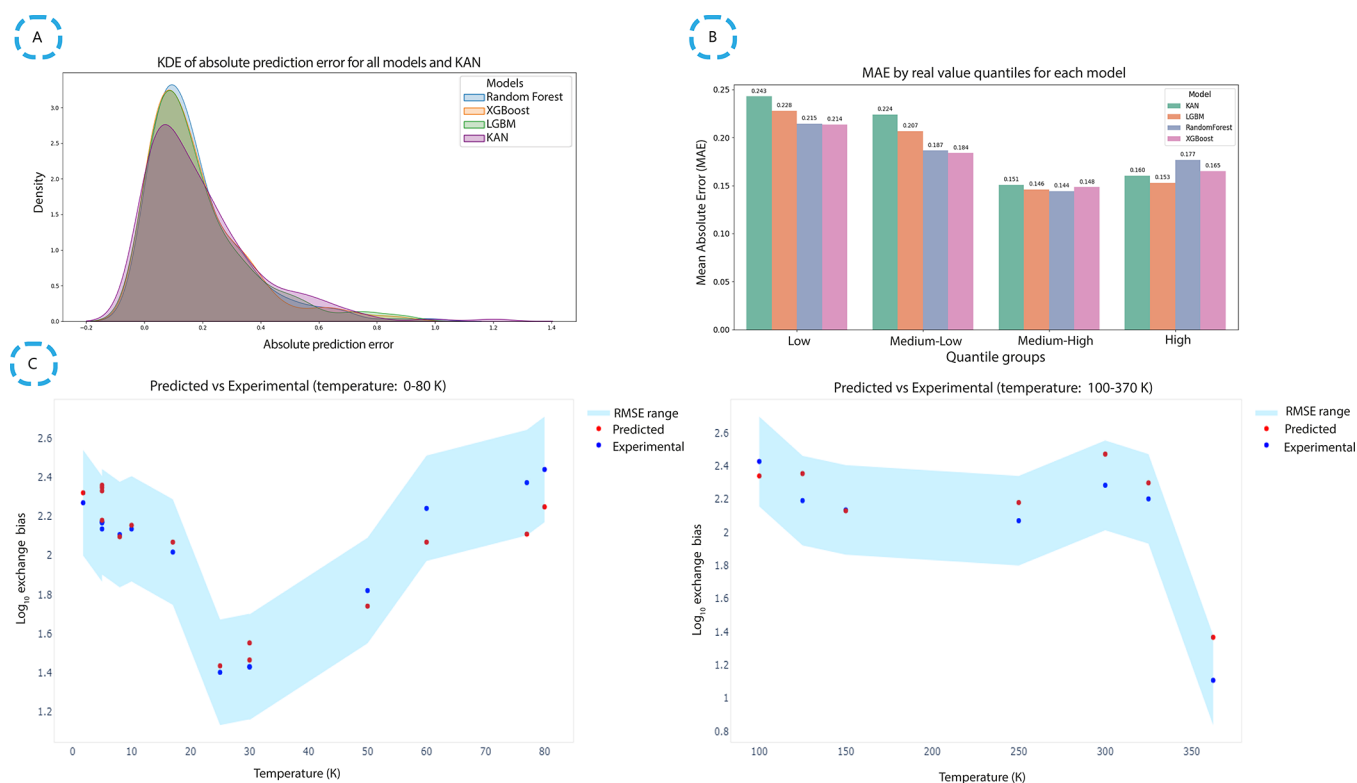


Figure 5. Additional evaluation of the validation set. (A) Distribution of absolute errors for all models and KAN; (B) mean absolute error (MAE) by quantile groups for models and KAN; (C) comparison of predicted versus true values of $\log_{10}$ EB across temperature ranges (0–80 and 100–370 K) for the XGB model. Shaded regions indicate the mean RMSE from CV, reflecting prediction uncertainty.

For the KAN network, key features for predicting EB include temperature parameters, nanoparticle geometric characteristics, and lattice parameters with core electronegativity being less significant. The lattice parameter plays the strongest role in influencing the EB, while geometric anisotropy and the core beta angle exhibit more complex nonlinear relationships. Other factors, such as the VEC difference and lattice parameters, show less significant impact on predictions of EB. KAN's balanced approach to decision-making, with more evenly distributed feature importance, offers a more uniform weighting of key factors compared to other models. This indicates that while KAN's generalization capability is similar to other ML models, it achieves more balanced decision-making across diverse parameters. Further analysis using the Mann–Whitney U test results indicates that the KAN model shows statistically significant differences in predicting EB for features like experimental temperature, magnetic field strength, maximum field range, and core b lattice parameter, with stronger significance observed in KAN for the latter two. These findings suggest that KAN captures important variations in these key features, which may contribute to its lower prediction error at higher EB values. For more detailed distributions and statistics, refer to [Supporting Information, Table S4 and Figure S8](#).

Detailed discussion on the role of temperature parameters, nanoparticle geometric characteristics, and lattice parameters in the KAN model, along with additional insights into SHAP values and the Mann–Whitney U test calculations that help identify the source of lower prediction errors in EB at higher values can be found in [Supporting Information, Sections 4 and 5](#).

To evaluate our models, we compared their predictions of logarithmic EB against experimental measurements in both the

test and validation data sets. We assessed how predictions varied with temperature to determine the model's ability to capture the temperature dependence of EB. This comprehensive evaluation showed that a significant proportion of validation samples fell within the CV RMSE range, indicating a high prediction accuracy.

We used a test data set to further assess model performance ([Table S5, Supporting Information](#)). This data set introduced extra variability through previously unseen feature combinations. A table of true versus predicted values and their differences is available in the [Supporting Information \(Table S6\)](#). [Figure 5A](#) shows that while all ML models have similar density and error distributions, KAN exhibits a lower density where errors are most concentrated, with a higher error rate in the 0.4 to 0.8 range compared to the other models. [Figure 5B](#) reveals that XGB is the most stable across all quantiles, but KAN performs better in predicting medium-high and high values. Specifically, XGB has absolute errors of 0.148 (medium-high) and 0.165 (high) compared to KAN's 0.151 and 0.16, respectively.

As an additional validation effort, we also analyzed the temperature dependence of the EB, as it is an important characteristic for understanding the stability and performance of magnetic systems. [Figure 5C](#) shows 20 data points. The strong agreement between real and predicted values, particularly in the critical 0 to 370 K temperature range for EB, highlights the model's accuracy in the prediction of the EB. Although not clearly visible, discrepancies between the predicted and experimental values of EB were lower in the 0–80 K range, likely due to fewer data points of the corresponding range in the collected database, as shown in [Figure 5B](#).

Despite high performance, our models were trained on a limited data set, which introduces inherent biases. The descriptors used might not capture all aspects of heterostructure nanoparticles, such as shell thickness or material ratios, leading to potential improvements in model predictions if more diverse data were available (Figure S2).

In conclusion, while our models are effective and interpretable, expanding the database and leveraging advanced deep learning techniques will further enhance the predictive accuracy and material design in the future.

CONCLUSIONS

In this study, we developed a novel predictive model for EB, achieving a validation R^2 score of 0.75 with the XGB model. This represents a significant advancement in simplifying the process of predicting EB values in heterostructure nanoparticles. By constructing a comprehensive data set and utilizing ML approaches, we have provided an effective solution to a critical problem in the field of material science. These ML predictions contribute directly to the design and optimization of materials with desirable EB properties by enabling rapid and precise evaluations of the material compositions and configurations. Models trained on data sets encompassing 37 features linked to EB behavior can predict material performance without resorting to computationally demanding DFT calculations, significantly accelerating the discovery process.

To validate our approach, we compared the performance of several ML models, including KAN and traditional computational simulations. While the XGB model demonstrated the highest predictive accuracy with a test R^2 of 0.74, the KAN model, despite its lower overall accuracy (R^2 of 0.67), excelled in capturing small, subtle variations in EB. This ability to model nuanced changes is essential for in-depth material analysis and the design of advanced magnetic materials. Furthermore, ML models can complement traditional computational methods by acting as an initial screening tool to identify the most promising materials for further investigation using high-fidelity techniques like micromagnetic simulations or *ab initio* calculations. In some cases, these models may even replace conventional approaches for routine predictions, particularly during exploratory phases.

Key parameters identified as influential in predicting EB included coercivity, anisotropy field, saturation magnetization, and the Néel-to-blocking temperature ratio. Our findings align with established literature but also highlight less frequently discussed factors such as Pauling electronegativity and the number of magnetic ions, significantly affecting EB. These lesser-studied factors provide new insights for optimizing magnetic materials and discovering novel behaviors in complex nanostructures.

Our analysis demonstrates that ML models, particularly XGB, streamline and enhance the prediction of magnetic properties in nanoparticles, offering robust predictive capabilities. However, KAN's strength in capturing significant EB reinforces the value of combining it with other ML methods for a more comprehensive understanding of magnetic behavior. Integrating these approaches with traditional computational techniques, such as micromagnetic simulations and *ab initio* calculations, will further advance material science, particularly in the development of materials tailored for electronics, spintronics, and magnetic storage applications. By providing deeper insights into physicochemical properties, such as

interlayer interactions, temperature dependencies, and magnetic susceptibility, these methods accelerate material discovery and optimization, enabling the development of advanced materials with tailored performance characteristics.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.4c07028>.

Database collection (Table S1 and Figure S1), data preprocessing and analysis (Figures S2–S5 and Table S2), ML models (Table S3 and Figure S6), SHAP analysis details (Figures S7 and S8 and Table S4), prediction results (Tables S5 and S6) (PDF). The data and the code supporting the findings of this study are openly available at <https://github.com/ai-chem/MagnetExchangeBias>

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

ABBREVIATIONS

EB, exchange bias; AFM, antiferromagnetic; FM, ferromagnet; FiM, ferrimagnet; SG, spin-glass; SD, spin disorder; PM, paramagnet; LGBM, LightGBM; DFT, density functional theory; KAN, Kolmogorov-Arnold Network; MLP, multilayer perceptron; SHAP, Shapley additive explanations; KDE, kernel density estimates; RMSE, root-mean-square error; MAE, mean absolute error.

REFERENCES

- (1) Gupta, R.; Kotnala, R. K. A review on current status and mechanisms of room-temperature magnetoelectric coupling in multiferroics for device applications. *J. Mater. Sci.* **2022**, *57* (27), 12710–12737.
- (2) Li, Z.; Yao, Z.; Haidry, A. A.; Luan, Y.; Chen, Y.; Zhang, B. Y.; Xu, K.; Deng, R.; Duc Hoa, N.; Zhou, J.; et al. Recent advances of atomically thin 2D heterostructures in sensing applications. *Nano Today* **2021**, *40*, No. 101287.
- (3) Mi, M.; Xiao, H.; Yu, L.; Zhang, Y.; Wang, Y.; Cao, Q.; Wang, Y. Two-dimensional magnetic materials for spintronic devices. *Mater. Today Nano* **2023**, *24* (September), No. 100408.
- (4) Raza, M. H.; Di Chio, R.; Movlaee, K.; Amsalem, P.; Koch, N.; Barsan, N.; Neri, G.; Pinna, N. Role of Heterojunctions of Core–Shell Heterostructures in Gas Sensing. *ACS Appl. Mater. Interfaces* **2022**, *14* (19), 22041–22052.
- (5) Phan, M. H.; Alonso, J.; Khurshid, H.; Lampen-Kelley, P.; Chandra, S.; Stojak Repa, K.; Nemati, Z.; Das, R.; Iglesias, O.; Srikanth, H. Exchange bias effects in iron oxide-based nanoparticle systems. *Nanomaterials* **2016**, *6* (11), 221.
- (6) Manna, P. K.; Yusuf, S. M. Two interface effects: Exchange bias and magnetic proximity. *Phys. Rep.* **2014**, *535* (2), 61–99.
- (7) Blachowicz, T.; Ehrmann, A. Exchange bias in thin films—an update. *Coatings* **2021**, *11* (2), 122–21.
- (8) Nogues, J.; Sort, J.; Langlais, V.; Skumryev, V.; Surinach, S.; Muñoz, J. S.; Baro, M. D. Exchange bias in nanostructures. *Phys. Rep.* **2005**, *422* (3), 65–117.
- (9) Feyngenson, M.; Bauer, J. C.; Gai, Z.; Marques, C.; Aronson, M. C.; Teng, X.; Su, D.; Stanic, V.; Urban, V. S.; Beyer, K. A.; Dai, S. Exchange bias effect in Au-Fe₃O₄ dumbbell nanoparticles induced by the charge transfer from gold. *Physical Review B - Condensed Matter and Materials Physics* **2015**, *92* (5), No. 054416.
- (10) Yadav, N.; Kumar, A.; Kumari, K.; Sharma, M. K.; Park, S.; Kumar, S.; Kim, J. W.; Yun, C.; Huh, S. H.; Koo, B. H. Exchange bias induced by CoMn₂O₄ interface grown through sequential annealing in Mn₂O₃@Co₃O₄ core–shell nanocomposite. *J. Am. Ceram. Soc.* **2024**, *1–15*.
- (11) Khalid, H.; Heo, S. G.; Lee, M. H.; Park, K. T.; Oh, S. J.; Seo, S. J. Versatile synthesis of 1-D akaganeite nanoparticles with fine-tuned exchange biasing and weak ferromagnetic effect. *J. Alloys Compd.* **2023**, *962*, No. 171056.
- (12) Lopez-Ortega, A.; Estrader, M.; Salazar-Alvarez, G.; Roca, A. G.; Nogues, J. Applications of exchange coupled bi-magnetic hard/soft and soft/hard magnetic core/shell nanoparticles. *Phys. Rep.* **2015**, *553*, 1–32.
- (13) Lavorato, G.; Lima, E.; Troiani, H.; Zysler, R.; Winkler, E. Tuning coercivity and exchange bias by controlling the interface coupling in bimagnetic core/shell nanoparticles. *Nanoscale* **2017**, *9*, 10240.
- (14) Lottini, E.; López-Ortega, A.; Bertoni, G.; Turner, S.; Meledina, M.; Van Tendeloo, G.; De Julián Fernández, C.; Sangregorio, C. Strongly Exchange Coupled Core/Shell Nanoparticles with High Magnetic Anisotropy: A Strategy toward Rare-Earth-Free Permanent Magnets. *Chem. Mater.* **2016**, *28* (12), 4214–4222.
- (15) Tsopoe, S. P.; Borgohain, C.; Fopase, R.; Pandey, L. M.; Borah, J. P. A comparative investigation of normal and inverted exchange bias effect for magnetic fluid hyperthermia applications. *Sci. Rep.* **2020**, *10* (1), 1–15.
- (16) Xing, M.; Mohapatra, J.; Elkins, J.; Guragain, D.; Mishra, S. R.; Ping Liu, J. Exchange bias and Verwey transition in Fe₅C₂/Fe₃O₄ core/shell nanoparticles. *Nanoscale* **2021**, *13* (37), 15837–15843.
- (17) Sartori, K.; Lopez-Martin, R.; Choueikani, F.; Gloter, A.; Grenèche, J. M.; Begin-Colin, S.; Taverna, D.; De Toro, J. A.; Pichon, B. P. Magnetic anisotropy engineering in onion-structured metal oxide nanoparticles combining dual exchange coupling and proximity effects. *Nanoscale Advances* **2024**, *6* (11), 2903–2918.
- (18) Banerjee, A.; Mondal, M. K.; De, D.; Goswami, S.; Das, S.; Chakrabarti, P. K. Exhibition of magnetic memory effects and exchange bias in nanocrystalline CoSb₂O₄. *Journal of Materials Chemistry C* **2024**, *12*, 19236.
- (19) Bohra, M.; Giaremis, S.; KS, A.; Mathioudaki, S.; Kioseoglou, J.; Grammatikopoulos, P. Ferromagnetic–Antiferromagnetic Coupling in Gas-Phase Synthesized M(Fe, Co, and Ni)–Cr Nanoparticles for Next-Generation Magnetic Applications. *Advanced Science* **2024**, *2403708*.
- (20) Giri, N.; Biswas, B. S.; Sengupta, P.; Ray, R. Exchange bias and magnetic memory effect in hole doped Nd_{0.78}Sr_{0.22}CoO₃ nanoparticles. *RSC Adv.* **2022**, *12* (41), 26470–26475.
- (21) Ge, J.; Chen, R. R.; Ren, X.; Liu, J.; Ong, S. J. H.; Xu, Z. J. Ferromagnetic–Antiferromagnetic Coupling Core–Shell Nanoparticles with Spin Conservation for Water Oxidation. *Adv. Mater.* **2021**, *33* (42), 1–8.
- (22) Schuller, I. K.; Morales, R.; Batlle, X.; Nowak, U.; Güntherodt, G. Role of the antiferromagnetic bulk spins in exchange bias. *J. Magn. Magn. Mater.* **2016**, *416*, 2–9.
- (23) Rivas-Murias, B.; Testa-Anta, M.; Skorikov, A. S.; Comesaña-Hermo, M.; Bals, S.; Salgueiriño, V. Interfaceless Exchange Bias in CoFe₂O₄ Nanocrystals. *Nano Lett.* **2023**, *23* (5), 1688–1695.
- (24) Ávila-Gutiérrez, M.; Moisset, A.; Ngo, A. T.; Costanzo, S.; Simon, G.; Colomban, P.; Petit, M.; Petit, C.; Lisiecki, I. Influence of the nanocrystallinity on exchange bias in Co/CoO core/shell nanoparticles. *Colloids Surf, A* **2023**, *676* (August), No. 132281.
- (25) González, J. A.; Andrés, J. P.; López Antón, R.; De Toro, J. A.; Normile, P. S.; Muniz, P.; Riveiro, J. M.; Nogués, J. Maximizing Exchange Bias in Co/CoO Core/Shell Nanoparticles by Lattice Matching between the Shell and the Embedding Matrix. *Chem. Mater.* **2017**, *29* (12), 5200–5206.
- (26) Stamps, R. L. Mechanisms for exchange bias. *J. Phys. D: Appl. Phys.* **2000**, *33* (23), R247.
- (27) Singh, V.; Patra, S.; Murugan, N. A.; Toncu, D. C.; Tiwari, A. Recent trends in computational tools and data-driven modeling for advanced materials. *Materials Advances* **2022**, *3*, 4069–4087.
- (28) Muzzi, B.; Albino, M.; Petrecca, M.; Innocenti, C.; de Julián Fernández, C.; Bertoni, G.; Ibarra, M. R.; Christensen, M.; Avdeev, M.; Marquina, C.; Sangregorio, C. Defect-Engineering by Solvent Mediated Mild Oxidation as a Tool to Induce Exchange Bias in Metal Doped Ferrites. *Small Methods* **2023**, *7* (11), 1–12.
- (29) Kurichenko, V. L.; Karpenkov, D. Y.; Degtyarenko, A. Y. Experimental and micromagnetic investigation of texture influence on magnetic properties of anisotropic Co/Co₃O₄ exchange-bias composites. *J. Magn. Magn. Mater.* **2023**, *565* (September 2022), No. 170232.
- (30) Brown, W. F. Virtues and weaknesses of the domain concept. *Rev. Mod. Phys.* **1945**, *17* (1), 15–19.
- (31) Vivek, S.; Kumar, Ajith S.; Lekha, C. S. C.; Nair, S. S. Exchange bias in BiFeO₃ and Bi_{0.9}La_{0.1}FeO₃ nanoparticles. *J. Phys. D: Appl. Phys.* **2021**, *54*, 125301.
- (32) Bhandari, C.; Nop, G. N.; Smith, J. D. H.; Paudyal, D. Accurate machine-learning predictions of coercivity in high-performance permanent magnets. *Physical Review Applied* **2024**, *22* (2), 1.
- (33) Austin, B. M.; Zubarev, D. Y.; Lester, W. A. Quantum monte carlo and related approaches. *Chem. Rev.* **2012**, *112* (1), 263–288.
- (34) Kim, P.; Serov, N.; Falchevskaya, A.; Shabalkin, I.; Dmitrenko, A.; Kladko, D.; Vinogradov, V. Quantifying the Efficacy of Magnetic Nanoparticles for MRI and Hyperthermia Applications via Machine Learning Methods. *Small* **2023**, *19* (48), 1–9.
- (35) Iglesias, O.; Labarta, A. Finite-size and surface effects in maghemite nanoparticles: Monte Carlo simulations. *Physical Review B - Condensed Matter and Materials Physics* **2001**, *63* (18), 1–11.
- (36) Dieny, B.; Prejbeanu, I. L.; Garello, K.; Gambardella, P.; Freitas, P.; Lehnndorff, R.; Raberg, W.; Ebels, U.; Demokritov, S. O.; Akerman, J.; et al. Opportunities and challenges for spintronics in the microelectronics industry. *Nature Electronics* **2020**, *3* (8), 446–459.
- (37) Cheenikundil, R.; Lu, Z.; Pereiro, M.; Delin, A.; Thonig, D. Stability and dynamics of magnetic skyrmions in FM/AFM heterostructures. *Phys. Rev. B* **2024**, *110*, No. 104430.

- (38) Wu, R.; Xue, M.; Maity, T.; Peng, Y.; Giri, S. K.; Tian, G.; Macmanus-Driscoll, J. L.; Yang, J. Influence of atomic roughness at the uncompensated Fe/CoO(111) interface on the exchange-bias effect. *Phys. Rev. B* **2020**, *101* (1), 14425.
- (39) Singh, P.; Del Rose, T.; Palasyuk, A.; Mudryk, Y. Physics-Informed Machine-Learning Prediction of Curie Temperatures and Its Promise for Guiding the Discovery of Functional Magnetic Materials. *Chem. Mater.* **2023**, *35* (16), 6304–6312.
- (40) Zeni, C.; Rossi, K.; Glielmo, A.; Baletto, F. On machine learning force fields for metallic nanoparticles. *Adv. Phys.: X* **2019**, *4* (1), No. 1654919.
- (41) Trostianchyn, A.; Izonin, I.; Tkachenko, R.; Duriagina, Z. (2022). Prediction of Magnetic Remanence of Sm-Co Magnets Using Machine Learning Algorithms. In: Hu, Z.; Petoukhov, S.; Yanovsky, F.; He, M. (eds) *Advances in Computer Science for Engineering and Manufacturing*. ISEM 2021. Lecture Notes in Networks and Systems, vol 463. Springer: Cham. .
- (42) Xin, C.; Yin, Y.; Song, B.; Fan, Z.; Song, Y.; Pan, F. Machine learning-accelerated discovery of novel 2D ferromagnetic materials with strong magnetization. *Chip* **2023**, *2* (4), No. 100071.
- (43) Quach, D. T.; Pham, D. T.; Ngo, D. T.; Phan, T. L.; Park, S. Y.; Lee, S. H.; Kim, D. H. Minor hysteresis patterns with a rounded/sharpened reversing behavior in ferromagnetic multilayer. *Sci. Rep.* **2018**, *8* (1), 1–9.
- (44) Leliaert, J.; Mulkers, J. Tomorrow's micromagnetic simulations. *J. Appl. Phys.* **2019**, *125*, (18)
- (45) Zhang, C.; Zhang, Z.; Wang, D.; Hu, Y. Strain-controlled Néel temperature and exchange bias enhancements in IrMn/CoFeB bilayers. *J. Appl. Phys.* **2024**, *124*, (8)
- (46) Kadim, G.; Masrour, R.; Jabar, A. Magnetocaloric, electronic, magnetic, optical and thermoelectric properties in antiferromagnetic semiconductor GdCrO₃: Monte Carlo simulation and density functional theory. *J. Crystal Growth* **2022**, *581*, No. 126509.
- (47) Bunyaev, S. A.; Budinska, B.; Sachser, R.; Wang, Q.; Levchenko, K.; Knauer, S.; Bondarenko, A. V.; Urbánek, M.; Guslienka, K. Y.; Chumak, A. V.; et al. Engineered magnetization and exchange stiffness in direct-write Co-Fe nanoelements. *Appl. Phys. Lett.* **2021**, *118* (2), No. 022408.
- (48) Lomonosov, A. M.; Temnov, V. V.; Wegrowe, J. E. Anatomy of inertial magnons in ferromagnetic nanostructures. *Phys. Rev. B* **2021**, *104* (5), 1–7.
- (49) Shim, J. H.; Syed, A. A.; Shin, Y.; Kim, J. W.; Piao, H. G.; Lee, S. H.; Lee, K. M.; Jeong, J. R.; Kim, D. H.; Kim, D. E. Ultrafast dynamics of exchange stiffness in Co/Pt multilayer. *Communications Physics* **2020**, *3* (1), 1–8.
- (50) Montiel, M. G. Magnetic Characterization of Nanoparticles. In *Nanochemistry: Synthesis, Characterization and Applications*, 1st ed; CRC Press, 2023, pp. 141–160. .
- (51) Dumitru, I.; Bulai, G.; Caltun, O. F. Magnetization processes in ferrite nanoparticles, thin films, and nanowires. In *Ferrite Nanostructured Magnetic Materials*. Woodhead Publishing, 2023; pp 55–70. .
- (52) SpringerMaterials Home Page. <https://materials.springer.com> (accessed 2024–09–25).
- (53) Grinsztajn, L.; Oyallon, E.; Varoquaux, G. Why do tree-based models still outperform deep learning on typical tabular data? *Adv. Neural Info. Process. Syst.* **2022**, *35*, 507–520.
- (54) Rana, P. S.; Kalpana, Chahat; Modi, S. K.; Yadav, A. L., Singla, S. (2023). Comparative Analysis of Tree-Based Models and Deep Learning Architectures for Tabular Data: Performance Disparities and Underlying Factors. *Proceedings - 2023 International Conference on Advanced Computing and Communication Technologies, ICACCTech 2023*, 224–231. .
- (55) Wang, R.; Yu, H.; Zhong, Y.; Xiang, H. Efficient prediction of potential energy surface and physical properties with Kolmogorov-Arnold Networks. *arXiv preprint arXiv:2409.03430* **2024**.
- (56) Wang, Y.; Sun, J.; Bai, J.; Anitescu, C.; Eshaghi, M. S.; Zhuang, X.; Rabczuk, T.; Liu, Y. Kolmogorov Arnold Informed neural network: A physics-informed deep learning framework for solving forward and inverse problems based on Kolmogorov Arnold Networks. *Computer Methods in Applied Mechanics and Engineering* **2025**, 433 (PB), No. 117518.
- (57) Samadi, M. E.; Müller, Y.; Schuppert, A. Smooth Kolmogorov Arnold networks enabling structural knowledge representation. *arXiv preprint arXiv:2405.11318*. **2024**.
- (58) O'Grady, K.; Fernandez-Outon, L. E.; Vallejo-Fernandez, G. A new paradigm for exchange bias in polycrystalline thin films. *J. Magn. Magn. Mater.* **2010**, *322* (8), 883–899.
- (59) Dobrynin, A. N.; Warin, P.; Vorobiev, A.; Givord, D. On the origin of positive exchange bias and coercivity enhancement in proximity to the blocking temperature. *J. Magn. Magn. Mater.* **2021**, *520* (March 2020), No. 166707.
- (60) Dimitrov, D.; Zhang, S. Effect of exchange interactions at antiferromagnetic/ferromagnetic interfaces on exchange bias and coercivity. *Phys. Rev. B Condens. Matter Mater. Phys.* **1998**, *58* (18), 12090–12094.
- (61) Hu, J. G.; Jin, G.; Hu, A.; Ma, Y. Q. Temperature dependence of exchange bias and coercivity in ferromagnetic/antiferromagnetic bilayers. *European Physical Journal B* **2004**, *40* (3), 265–271.
- (62) Sivarajani, K. S.; Antilen Jacob, G.; Justin Joseyphus, R. Coercivity and exchange bias in size reduced iron obtained through chemical reduction. *J. Magn. Magn. Mater.* **2020**, *513* (April), No. 167228.
- (63) Sterwerf, C.; Meinert, M.; Arenholz, E.; Schmalhorst, J. M.; Reiss, G. Room Temperature Exchange Bias in BiFeO₃/Co-Fe Bilayers. *IEEE Trans. Magn.* **2016**, *52* (7), 1–21.
- (64) Brück, S.; Sort, J.; Baltz, V.; Suriñach, S.; Muñoz, J. S.; Dieny, B.; Baró, M. D.; Nogués, J. Exploiting length scales of exchange-bias systems to fully tailor double-shifted hysteresis loops. *Adv. Mater.* **2005**, *17* (24), 2978–2983.
- (65) Tracy, J. B.; Weiss, D. N.; Dinega, D. P.; Bawendi, M. G. Exchange biasing and magnetic properties of partially and fully oxidized colloidal cobalt nanoparticles. *Phys. Rev. B Condens. Matter Mater. Phys.* **2005**, *72* (6), No. 064404.
- (66) Xiang, H. J.; Whangbo, M. H. Cooperative effect of electron correlation and spin-orbit coupling on the electronic and magnetic properties of Ba₂NaOsO₆. *Physical Review B - Condensed Matter and Materials Physics* **2007**, *75* (5), 1–4.
- (67) Smith, D. W. A new approach to the relationship between bond energy and electronegativity. *Polyhedron* **2007**, *26* (2), 519–523.
- (68) Smith, D. W. Effects of exchange energy and spin-orbit coupling on bond energies. *J. Chem. Educ.* **2004**, *81* (6), 886–890.
- (69) Iglesias, Ò.; Labarta, A.; Batlle, X. Exchange bias phenomenology and models of core/shell nanoparticles. *J. Nanosci. Nanotechnol.* **2008**, *8* (6), 2761–2780.
- (70) Lemos, C. G. O.; Figueiredo, W.; Santos, M. Exchange bias for core/shell magnetic nanoparticles. *Physica A: Statistical Mechanics and Its Applications* **2015**, *433*, 148–160.